Javad Bajelan

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Education

Azarbaijan Shahid Madani University Sep. 2022 – Aug. 2026
Bachelor of Science in Computer Science (B.Sc.), **GPA: 17.91/20** Tabriz, Iran

Courses: AI (20/20), Data Mining (20/20), Information Retrieval (19.2/20), Compiler Design (20/20), Operating Systems (20/20), Database Systems (20/20), Algorithms and Data Structures (20/20), Theory of Computation (20/20), Linear Algebra (20/20), Abstract Algebra (20/20), Computer Systems (20/20)

Bachelor's Research Project (Supervisor: Prof. Jalil Ghavidel Neycharan):

- Project Title: Persian Legal LLM Development with OCR, RAG, and LoRA Fine-Tuning. GitHub

Working Experience

- **Innovatex – United Arab Emirates** Mar 2025 – Present
  Go Backend Developer (innovatex.ae)
  - **IXCbill billing & cloud (ixnodes)** — Go billing engine with CloudStack VPS lifecycle management: invoice & renewal lifecycle (reopen expired/void payment windows, recover reusable provider attempts, re-issue renewals), multi-rail payments with automatic fallback of double bank payments to customer credit, per-jurisdiction VAT control, and a revenue dashboard (all-time/24h KPIs, credit-drawdown deduplicated); VPS lifecycle orchestration against CloudStack 4.20: create/destroy/expunge with 10-retry/10-min backoff and admin re-provision, live VM state sync, bandwidth metering (bytes-accurate, per-VM scoped), hostname auto-numbering, SSH-key/password bootstrap.
  - **TirPay payment suite** — 1,000+ commits: Go backend for multi-currency payments and purchases (website payments, gift cards, crypto/fiat exchange, credit cards) with a polling aggregator for crypto and world fiat rates, plus a provider-agnostic credit-card platform: virtual/physical card issuance, card & transaction ledgers via IBS, wallet management and crypto deposit management via GoWallet, IRR deposit via ZariPal.
  - **Innovatex BaaS Suite** — provider-agnostic white-label card API (Papara, Alaan, Mamo adapters) with a Go SDK (HMAC request signing), admin panel and signed mail-forwarding service; hardened security, regression tests and per-service data scoping.
  - **GoWallet** — multi-network crypto wallet API: hot/fee-wallet and token CRUD, funds-collection flow and transaction monitoring, ipn, hardened security, ops.
- **Meshtek – Turkey** Jul 2024 – Mar 2025
  Go/Python Backend Developer (meshtek.com.tr)
  - **Papara payment platform** — Go payment gateway for msamex exchange + Django admin panel (139 commits): card/atm operations, order flow, Telegram audit notification.
  - **MSAMEX exchange Telegram bot** — limit/market orders with order-book sorting and open-order filters, deposit-address flow and SQLite persistence, plus an integrated Xray VPN service (vmess/vless config issuance, account sync, expiry-based deactivation, traffic accounting).
  - **Market Maker** — Go pure market maker strategy for Multi-Exchanges (Binance, Bybit, Mexc, ...) with pricing rule (configurable spreads per pair), trade-repair tooling, gRPC/protobuf plumbing and commands, distributing bot instances across servers and central control panel, ops.
- **ShayTeck – Iran** Feb 2024 – Apr 2024
  Remote Python Developer (shayteck.com)
  - Python relay bot pushing tweets, retweets, quotes and media from tracked accounts to Telegram channels on a admin panel, session/cookie handling, Postgres backend.

  - Pyrogram bot to create and revoke chat invite links with live notifications, admin management and `openpyxl` xlsx reports; Dockerized and Postgres backend.
  - Persian-language auto-responder bot mapping keywords to canned text/media-group answers published to a target channel, with admin UI and JSON-backed storage.

- **Gentoo – OpenSource** Nov 2022 – Jan 2025
  Clang/Musl Profile Maintainer (clang-musl-overlay – gentoo-patchset)
  - 125+ commits (∼37% of the repo) packaging LLVM/Clang, Rust, Node.js, GHC and SBCL for a musl-based toolchain.
  - Release engineering across 10+ toolchain packages, tracking upstream versions to keep the overlay current with LLVM 17/18, Rust 1.70–1.80 and Node.js 20/22.
  - Authored Clang/musl portability patches for gcr, valgrind, ethtool, sshfs, ffcall and MIT Scheme (type mismatches, `-lgcc` link errors).

Academic Projects

- **The Dolme Programming Language** – GitHub
  - Built a complete compiler in Go for a type-checked language (typed functions, recursion, `if`/`while`, break/continue): lexer, LL(1) parser with explicit parsing table, syntax-directed translation to a three-address-code IR, and an IR interpreter.
  - Added semantic checks – declaration type validation, call arity/type checking, bool-typed operators – with backpatching to handle returns and branching in code generation.
  - Wrote an ARM64 macOS assembly backend (large constants via `ldr` pseudo-instructions) and a differential integration suite: all 19 example programs (Taylor-series sine, Newton sqrt, primality, Fibonacci) run both interpreted and as native binaries with outputs compared against expected results.
- **LOB Sequence Classification** – GitHub.
  - Built the training & evaluation harness for mid-price direction prediction on a public LOB dataset: 144 features, five horizons (10–100 ticks), sliding-window sequences over pre-split walk-forward folds.
  - Trained and cross-validated MLP/CNN/LSTM/TCN baselines against tensorized (TABL) and attention-based (TLOB) models, reporting mean±std macro-F1 per horizon (up to 0.80) with confusion-matrix analysis.
  - Refactored the data pipeline (LZMA-compressed matrices → typed NumPy loaders, centralized preprocessing/model defaults) and instrumented effective-attention extraction to inspect what the models actually attend to.
- **RISC-V Libc** – GitHub
  - Building an experimental C standard library for RISC-V 64 with low-level libc routines.
  - Uses a cross-compilation workflow with RISC-V tooling and emulator-based validation.

Teaching Experience

- **Teaching Assistant** at Azarbaijan Shahid Madani University 2023-2025
  - Fundamentals of Logic — Supervisor: Prof. Somayyeh Tari
  - Computation Theory — Supervisor: Prof. Somayyeh Tari
  - Linear Algebra — Supervisor: Prof. Somayyeh Tari

Skills

- **Programming Languages:** Go, Python, C/C++, Rust, x86/RISC-V/ARM64 Assembly, Coq/Rocq, Haskell

- **Frameworks / Libraries:** Go (Kratos, GORM, Gin, Echo, Fiber, gRPC); Python (Django, Flask, Pydantic, PyTorch, Transformers, Scikit-learn, NLTK, Scipy, Pandas, Numpy, Scapy); C (OpenMP, OpenSSL, pthreads, libcurl, libuv)
- **Developer / Debugging Tools:** LLVM/Clang, GCC, GDB, LLDB, ASan, radare2, Ghidra, GNU Binutils (objdump, readelf), strace, ltrace, Valgrind, perf, Wireshark, Nginx, Git, Latex
- **Backend / Systems:** Microservices, Linux/Unix Systems Programming, POSIX APIs, Processes & Threads, Concurrency & Synchronization, Socket Programming, TCP/IP, Distributed Systems, REST/gRPC Services, Performance Analysis, Memory Debugging, Secure Coding, Load Balancing, Caching, Docker, CI/CD, GitHub Actions, Linux Administration, Infrastructure as Code (IaC), Firewalls
- **Algorithms / ML:** Algorithm Development, Deep Learning, Transformer Architectures, Retrieval-Augmented Generation, LoRA/PEFT Fine-Tuning, Search Engines, Embeddings & Vector Search
- **Theoretical:** First-Order Logic, Mathematical Logic, Computability & Complexity Theory, Type Theory, Category Theory, Lambda Calculus, Proof Theory, Denotational & Operational Semantics, Axiomatic Semantics, Intuitionistic Logic, Relational Model, Relational Algebra & Calculus, Schema Design & Normalization
- **Languages:** English, Persian